

CE 1201 · ENGINEERING MECHANICS · KUET / BECM

Formula Book + Theory Bank + Proof & Derivation Manual

8-Set Exam System · Regular Final Focus · বাংলা Concept Support

A1
Centroid

A2
Area MOI

A3
POI · Mass MOI ·
Pappus

A4
Dynamics

B1
Force · FBD

B2
SFBM · Pulley

B3
Truss · Motion

B4
Friction · Cable

এই PDF কীভাবে পড়বেন

এটি শুধু formula sheet নয়। প্রতিটি set-এ **Formula → Theory → QB pattern → Proof/Derivation** ধারায় সাজানো হয়েছে, যাতে পরীক্ষায় ৪ সেটের মধ্যে অন্তত ৩ সেট confidently answer করা যায়।

3-out-of-4 Rule

প্রতি section-এ ৪টি full-question set থাকে; উত্তর দিতে হয় যেকোনো ৩টি। তাই minimum strategy: **৩টি set full + ১টি backup theory/formula.**

2020 question pattern-কে supplied instruction অনুযায়ী exception হিসেবে ধরা হয়েছে।

Priority Code

VERY HIGH = repeated / core anchor question

HIGH = frequent and structurally important

MEDIUM = supportive / mixed-year topic

QB year tags শুধু uploaded notes/papers-এ স্পষ্টভাবে চিহ্নিত pattern থেকে ব্যবহার করা হয়েছে।

8-Set Map

Part	Set	Core topics	Must-have proof / skill
Section A	A1	Centroid, CG, CM, symmetry, composite area	Triangle centroid; sector/arc location
	A2	Area MOI, polar MOI, radius of gyration	$J = I_x + I_y$; parallel-axis; rectangle/triangle MOI
	A3	Product of inertia, Mass MOI, Pappus	POI transfer; standard mass MOI; Pappus theorems
	A4	Kinematics/Kinetics, Work-Energy, Impulse-Momentum	$W_{net} = \Delta KE$; $R\Delta t = \Delta(mv)$; projectile equation
Section B	B1	Force system, equilibrium, FBD, supports	Component/resultant method; correct FBD
	B2	Shear force, bending moment, pulley	Load equivalence; beam section method
	B3	Truss, plane/rectilinear motion	Method of joints/sections; projectile motion
	B4	Friction, wedge, belt friction, flexible cord	Belt friction; parabola/catenary equations

Source integrity / শুদ্ধতা নীতি

Main terminology and course organization follow the uploaded notes/slides/QB. Where a prepared note contains an obvious notation or unit inconsistency, the manual does not silently copy it; the discrepancy is flagged and the dimensionally consistent notation is used.

Symbols, Units & Exam Writing Rules

Core symbols

Item	Symbol / relation	Dimension / unit
Centroid	$\bar{x}, \bar{y}$	geometric location
Area	A	L^2
Area MOI	I_x, I_y, J	L^4
Radius of gyration	$k = \sqrt{I/A}$	L
Mass MOI	$I = \int r^2 dm$	$M \cdot L^2$
Force / resultant	F, R	N, lb
Moment	$M = Fd_{\perp}$	$N \cdot m, lb \cdot ft$
Impulse	$\int F dt$	$N \cdot s$
Momentum	mv	$N \cdot s$

Exam consistency rules

- Angle reference axis আগে লিখুন: θ যদি +x axis থেকে হয়, তবে $F_x = F \cos \theta$, $F_y = F \sin \theta$.
- Area MOI unit = mm^4 / in^4 ; mass MOI = $kg \cdot m^2 / slug \cdot ft^2$.
- Radius of gyration **length**; unit কখনো in^4 নয়.
- Statics-এ equilibrium: $\Sigma F_x = 0$, $\Sigma F_y = 0$, $\Sigma M = 0$.
- Hole/cut-out: composite area এবং MOI table-এ negative contribution.
- Friction: static friction সবসময় μN নয়; limiting অবস্থায় $F = \mu N$.
- Pulley: same continuous ideal rope-এ tension T constant; $W/2/W/4$ shortcut universal নয়.

Important correction

Product-of-inertia prepared solution page prints a $length^4$ unit beside radius of gyration. The formula $k = \sqrt{I/A}$ makes k a length; this manual uses the correct length unit while retaining the source formula pattern.

Answer presentation shortcut

1. Sketch / FBD

যেখানে force/geometry আছে, first clean figure.

2. Formula + condition

কোন theorem/assumption ব্যবহার করছেন লিখুন.

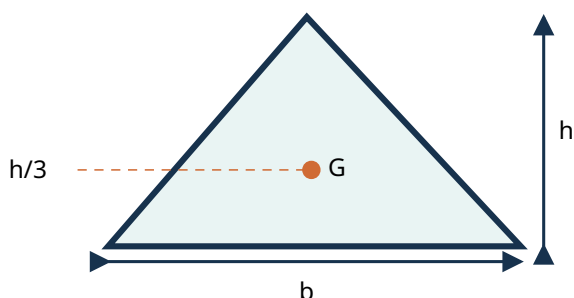
3. Unit + sense

Final answer-এর unit, direction, T/C, sign interpretation দিন.

A1

Centroid · Center of Gravity · Center of Mass

Repeated anchor set | geometry-first thinking



Bangla concept

Centroid = কোনো geometrical area/line/ volume-এর ভারসাম্য-ভিত্তিক জ্যামিতিক কেন্দ্র। Uniform density body হলে centroid ও center of mass একই স্থানে আসে।

Core integral formulas

$$\bar{x} = (\int x \, dA)/A \quad ; \quad \bar{y} = (\int y \, dA)/A$$

$$\bar{x} = \Sigma(A_i x_i)/\Sigma A_i \quad ; \quad \bar{y} = \Sigma(A_i y_i)/\Sigma A_i$$

Centroid Formula Bank

Quantity / Case	Formula	Note / Unit
Rectangle $b \times h$	$\bar{x}=b/2, \bar{y}=h/2$	from a corner
Triangle	$\bar{y}=h/3$	from base; $\bar{x}=b/3$ for right triangle measured from the corresponding side
Semicircular area	$\bar{y}=4r/(3\pi)$	from diameter along symmetry axis
Quarter-circular area	$\bar{x}=\bar{y}=4r/(3\pi)$	from straight sides
Circular sector, angle 2α	$\bar{x}=2r \sin\alpha/(3\alpha)$	along symmetry axis; α in radians
Circular arc, angle 2α	$\bar{x}=r \sin\alpha/\alpha$	line/arc centroid; α in radians
Composite area	$\bar{x}=\Sigma A_i x_i/\Sigma A_i$	take holes as negative A

High-priority Theory + QB

VERY HIGH**Define centroid.**

Tagged in prepared note: 2015-2022

HIGH**Real-life applications of centroid.**

Tagged: 2016, 2019, 2021

HIGH**Describe axis of symmetry with figure.**

Tagged: 2017, 2018, 2020

HIGH**Find centroid of a triangle of base b and altitude h .**

Tagged: 2016, 2018, 2021

HIGH**Locate centroid of circular sector of radius r , central angle 2β .**

Tagged: 2017, 2019

MEDIUM**Define center of gravity; distinguish centroid and CG/CM.**

CG tagged: 2019, 2020, 2022

A1 THEORY BANK

Exam-Ready Short Theory - Centroid

Centroid

Definition: The centroid is the geometrical centre of a line, area or volume. For an area, its coordinates are obtained from the first moments of area: $\bar{x}A = \int x \, dA$ and $\bar{y}A = \int y \, dA$.

বাংলায়: shape-টির ক্ষুদ্র ক্ষুদ্র area element-এর অবস্থানকে area দিয়ে weighted average করলে যে বিন্দু পাওয়া যায় সেটিই centroid.

Axis of Symmetry

যে axis-এর এক পাশে থাকা প্রতিটি elementary area-এর বিপরীতে অপর পাশে সমান corresponding area থাকে, সেই axis হলো symmetry axis. একটি area-এর একটি symmetry axis থাকলে centroid সেই axis-এর উপর; দুটি symmetry axis থাকলে centroid তাদের intersection-এ.

Centroid vs Center of Gravity

Centroid	Center of Gravity
Geometrical property	Resultant weight acts through this point
Line/area/volume geometry	Physical body + gravitational field
Uniform density body-তে CM-এর সাথে coincide করতে পারে	Uniform gravitational field-এ CM-এর সাথে coincide করে

Fast composite-area technique

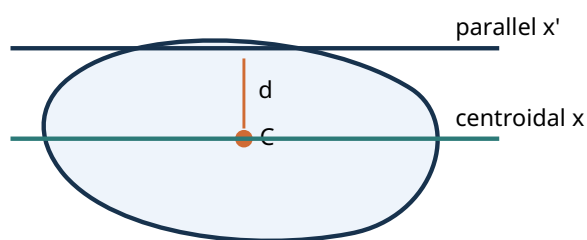
1. Reference x-y axes ঠিক করুন.
2. Shape-কে simple parts-এ ভাগ করুন.
3. A_i , x_i , y_i table করুন.
4. Hole হলে A negative.
5. $\sum A_i x_i / \sum A_i$ এবং $\sum A_i y_i / \sum A_i$.

Memory trick

"Area × Distance, then divide by Total Area" = composite centroid. Symmetry থাকলে আগে symmetry ব্যবহার করুন - calculation অর্ধেক হয়ে যায়.

Core definitions

Quantity / Case	Formula	Note / Unit
About x-axis	$I_x = \int y^2 dA$	distance y from x-axis
About y-axis	$I_y = \int x^2 dA$	distance x from y-axis
Polar about z through same point	$J = \int r^2 dA$	$r^2 = x^2 + y^2$
Perpendicular-axis relation	$J = I_x + I_y$	plane area; axes meet at same point
Radius of gyration	$k_x = \sqrt{I_x/A}$, $k_y = \sqrt{I_y/A}$, $k_p = \sqrt{J/A}$	unit = length
Parallel-axis theorem	$I = \bar{I} + Ad^2$	d = separation of parallel axes
Radius transfer	$k^2 = \bar{k}^2 + d^2$	from $I = k^2A$



Bangla concept

Second moment of area মূলত বলে area-টি reference axis থেকে কতটা “ছড়িয়ে” আছে। একই area যত দূরে spread হবে, I তত বড় হবে; তাই beam bending/deflection analysis-এ এটি গুরুত্বপূর্ণ।

Standard Area MOI

Quantity / Case	Formula	Note / Unit
Rectangle $b \times h$, centroid x-axis	$I_x = bh^3/12$	axis parallel to b
Rectangle $b \times h$, centroid y-axis	$I_y = hb^3/12$	axis parallel to h
Triangle $b \times h$, about base	$I_{\text{base}} = bh^3/12$	base axis
Triangle $b \times h$, centroidal axis $\parallel$ base	$\bar{I} = bh^3/36$	parallel-axis from base
Circle radius r, centroid diameter	$I = \pi r^4/4$	either x or y diameter
Circle radius r, polar at center	$J = \pi r^4/2$	$J = I_x + I_y$

High-priority Theory / Proof

VERY HIGH

Define rectangular MOI, polar MOI and radius of gyration; prove $J = I_x + I_y$.

Recurring in regular papers; explicitly seen in 2018 and older paper

VERY HIGH

Prove parallel-axis theorem $I = \bar{I} + Ad^2$ (and hence $k^2 = \bar{k}^2 + d^2$).

Explicit QB pattern

HIGH

Applications/significance of first and second moment of area.

Recurring theory + numerical

HIGH

Derive MOI of rectangle about centroidal axis.

Lecture slide derivation

HIGH

Derive MOI of triangle about base / centroidal parallel axis.

Lecture slide derivation

A2 THEORY BANK**Moment of Inertia - Concepts & Common Traps****Rectangular vs Polar MOI**

Rectangular MOI is about an axis lying in the plane of area. **Polar MOI** is about an axis perpendicular to the plane. Both are geometrical properties and have unit L^4 .

Radius of Gyration

যদি পুরো area A -কে axis থেকে k দূরে concentrate করলে একই moment of inertia পাওয়া যায়, সেই k হলো radius of gyration. তাই $I = Ak^2$.

Common error

Area MOI এবং Mass MOI এক নয়: area MOI-তে dA এবং unit L^4 ; mass MOI-তে dm এবং unit $M \cdot L^2$. একই "I" symbol ব্যবহৃত হলেও context অবশ্যই লিখবেন.

Composite MOI Quick Algorithm

1. Centroidal/reference axis identify করুন.
2. Each part-এর own centroidal MOI formula লিখুন.
3. Required axis shifted হলে $+A_id_i^2$.
4. Hole/cut-out: পুরো term negative.
5. I_x এবং I_y আলাদা column-এ sum করুন.
6. Polar চাইলে same point-এ $J = I_x + I_y$.

Product of Inertia (POI)

Quantity / Case	Formula	Note / Unit
Definition	$P_{xy} = \int xy \, dA$	may be +, -, or 0
Symmetry	$P_{xy} = 0$	if x or y is an axis of symmetry
Transfer formula	$P_{xy} = \bar{P}_{xy} + A d_x d_y$	signed d_x, d_y
Average MOI	$I_{avg} = (I_x + I_y)/2$	principal calculation
Radius in Mohr-type relation	$R = \sqrt{\{[(I_x - I_y)/2]^2 + P_{xy}^2\}}$	source notation
Principal extremes	$I_{max} = I_{avg} + R; I_{min} = I_{avg} - R$	then $k = \sqrt{I/A}$

POI sign shortcut

xy product-এর sign quadrant-এর উপর নির্ভর করে.
 QI & QIII -> positive xy; QII & QIV -> negative xy.
 Symmetry থাকলে opposite contributions cancel.

Repeated Z-section pattern

Prepared POI note tags the Z-section product/
 principal inertia problem across **2017-2022**. তাই
 A3-তে এটি high-value numerical archetype.

Mass Moment of Inertia

Quantity / Case	Formula	Note / Unit
Definition	$I = \int r^2 \, dm$	mass distribution
Transfer / parallel axis	$I = \bar{I} + md^2$	mass version
Slender rod, through centre perpendicular to rod	$I_G = mL^2/12$	standard shape relation shown in source table
Slender rod, through end perpendicular to rod	$I_{end} = mL^2/3$	parallel-axis result
Solid sphere, any diameter	$I = 2mr^2/5$	source table
Solid cylinder, symmetry axis	$I = mr^2/2$	source table
Solid cylinder, centroid transverse axis	$I = m(3r^2 + L^2)/12$	source table
Weight to mass (US customary)	$m = W/g$	m in slug if W in lb, g in ft/s ²

A3 PAPPUS-GULDINUS

Pappus-Guldinus Formula & Theory

First theorem - Surface area

A plane curve of length L revolving about a fixed external axis generates a surface.

$$S = 2\pi \bar{r} L$$

$\bar{r}$ = distance of the curve centroid from axis.

Second theorem - Volume

A plane area A revolving about a fixed external axis generates a solid.

$$V = 2\pi \bar{r} A$$

$\bar{r}$ = distance of area centroid from axis.

Quantity / Case	Formula	Note / Unit
Cone surface (lateral)	$S = \pi r l$	generating line l , centroid distance $r/2$
Cylinder lateral surface	$S = 2\pi r h$	generating line h at r
Sphere surface	$S = 4\pi r^2$	semicircular generating arc
Cone volume	$V = \pi r^2 h / 3$	triangular generating area
Cylinder volume	$V = \pi r^2 h$	rectangular generating area
Sphere volume	$V = 4\pi r^3 / 3$	semicircular generating area
Torus (circle radius a , centroid radius b)	$V = 2\pi^2 a^2 b$	$V = A(2\pi b)$

Theory / QB targets

VERY HIGH

State both Pappus-Guldinus theorems and apply to surface/volume of revolution.

Explicit regular-exam pattern; 2018 includes pulley rim application

HIGH

Use Pappus to find cone painting surface / cone volume.

Lecture-slide worked pattern

HIGH

Pulley rim mass and weight by Pappus + density.

Lecture-slide / QB pattern

HIGH

Mass MOI of composite mallet / frustum with cylindrical hole.

Set-wise note + regular papers

HIGH

Define and explain applications of product of inertia.

Prepared POI note

Kinematics Formula Bank (constant acceleration)

Quantity / Case	Formula	Note / Unit
Velocity-time	$v = u + at$	constant a
Displacement-time	$s = ut + \frac{1}{2}at^2$	constant a
Velocity-position	$v^2 = u^2 + 2as$	constant a
Average velocity	$s = \frac{1}{2}(u+v)t$	constant a
Projectile x	$x = u \cos\theta \cdot t$	no air resistance
Projectile y	$y = u \sin\theta \cdot t - \frac{1}{2}gt^2$	upward +
Trajectory	$y = x \tan\theta - gx^2/(2u^2\cos^2\theta)$	eliminate t

Kinetics / Work / Energy

Quantity / Case	Formula	Note / Unit
Newton	$\Sigma F = ma$	translation
Work	$U = \int F \cdot ds$	constant collinear: $U = Fs$
Weight work	$U_W = W \cdot \Delta h$ with sign from motion	or $-\Delta PE$
Spring work	$U_s = \frac{1}{2}k(x_1^2 - x_2^2)$	work done by spring
Kinetic energy	$KE = \frac{1}{2}mv^2$	translation
Rotational KE	$KE_{rot} = \frac{1}{2}I\omega^2$	rigid-body rotation
Work-energy principle	$U_{net} = T_2 - T_1 = \Delta KE$	system of forces
Power	$P = dU/dt = F \cdot v$	instantaneous

Impulse / Momentum

Quantity / Case	Formula	Note / Unit
Linear impulse	$J = \int F dt$	constant F: $F\Delta t$
Linear momentum	$p = mv$	vector
Impulse-momentum	$\int \Sigma F dt = mv_2 - mv_1$	particle / CM translation
Angular impulse	$\int M_O dt$	about O
Angular impulse-momentum	$\int M_O dt = H_{O2} - H_{O1}$	fixed axis: $= I_O(\omega_2 - \omega_1)$
Coefficient of restitution	$e = \text{speed of separation} / \text{speed of approach}$	along line of impact

A4 THEORY & METHOD SELECTOR

Dynamics - Which Method Should I Use?

Given / Asked	Best starting method	Why
Force + time, need velocity/ displacement	$\Sigma F=ma$ + kinematics	Acceleration can be found directly
Force/displacement or spring/gravity, need speed	Work-Energy	Time is often eliminated
Force-time / short impact / jet, need velocity change	Impulse-Momentum	Direct link between impulse and $\Delta(mv)$
Projectile gap/jump	Kinematics in x and y	x gives t, y gives height
Rolling body	Work-Energy with translation + rotation	$KE = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$

Classical Dynamics

Supplied dynamics note divides classical dynamics into **Kinematics** (geometry of motion) and **Kinetics** (motion together with force/cause). Kinetics is then solved by force-mass acceleration, work-energy, or impulse-momentum methods.

Projectile / motorcycle shortcut

Horizontal distance আগে use করে time বের করুন: $t=x/(u \cos\theta)$. তারপর $y=u \sin\theta \cdot t - \frac{1}{2}gt^2$. Reference elevations carefully add/subtract করুন.

High-priority Theory / Proof

VERY HIGH

State and prove the principle of linear impulse and momentum.

Repeated regular-paper proof

VERY HIGH

Define work/power and prove $W_{net} = \Delta KE$.

Explicit older regular-paper proof

HIGH

Explain work and kinetic energy; solve inclined motion / spring work.

Recurring

HIGH

Motorcycle/projectile gap problem.

Tagged in prepared dynamics note: 2015, 2018, 2021

HIGH

Jet/blade using impulse-momentum components.

Recurring exam archetype

MEDIUM

Angular impulse and angular momentum.

Included in supplied book-note pages



Roller: 1 reaction

Pin: A_x , A_y Fixed: A_x , A_y , M **FBD shortcut**

Object-কে environment থেকে “কেটে আলাদা” করুন;
তারপর শুধু external forces/reactions দেখান. Rope/
cable $\rightarrow$ tension; smooth contact $\rightarrow$ normal
reaction; weight $\rightarrow$ CG দিয়ে downward.

Core Formula Bank

Quantity / Case	Formula	Note / Unit
Components (θ from $+x$)	$F_x = F \cos \theta$; $F_y = F \sin \theta$	sign by quadrant
Resultant	$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$	magnitude
Direction	$\theta_R = \text{atan2}(\sum F_y, \sum F_x)$	quadrant-aware
Moment of a force	$M = F \, d_{\perp}$	perpendicular distance
Couple	$M_c = F \, d$	two equal opposite parallel forces
Equilibrium	$\sum F_x = 0$; $\sum F_y = 0$; $\sum M = 0$	2D rigid body
Lami (3 concurrent forces)	$F_1/\sin \alpha = F_2/\sin \beta = F_3/\sin \gamma$	only 3-force equilibrium

Support Reactions

Support	Unknown reactions in 2D	Memory
Roller / smooth surface	One reaction normal to surface	Cannot resist tangent movement
Pin / hinge	A_x and A_y	Allows rotation, blocks translation
Fixed	A_x , A_y and M_A	Blocks translation + rotation

Repeated Theory / QB

VERY HIGH**Define FBD, transmissibility, two-force member, equilibrium.**

Prepared note tags: FBD 2015,16,17,18,21,22; two-force 16,17,18,21,22

HIGH**Draw FBD of all members of a structure.**

Repeated; recent 2023-2025 papers also use this style

HIGH**Illustrate support systems and reactions.**

Recent regular-paper theory

HIGH**Resultant of a coplanar concurrent force system.**

Core B1 numerical

HIGH**Wheel just rolling over a block - least force / reaction.**

Repeated archetype

B1 THEORY BANK

Definitions You Should Write Precisely

Free Body Diagram (FBD)

A diagram of an isolated body or group of bodies showing all external forces, reactions, applied loads, weight and friction acting on it.

Principle of Transmissibility

For a rigid body, replacing a force by another force of the same magnitude and direction anywhere along the **same line of action** does not change the external equilibrium or motion of the body.

Two-Force Member

If a member in equilibrium is acted upon by only two forces, those forces must be **equal, opposite and collinear**. A cable/rope is normally tension; a weightless boom may act in compression depending on the configuration.

Exam warning

Support reaction ভুল হলে পুরো statics numerical ভুল হবে। Figure দেখে support type identify না করে equation শুরু করবেন না.

B2

Shear Force · Bending Moment · Pulley

Equivalent loads + equilibrium + section cut



Resultant at $L/3$ from larger end

Distributed Load Equivalents

Quantity / Case	Formula	Note / Unit
Point load P	$R=P$	acts at given point
UDL w over L	$R=wL$	at $L/2$
Triangular UVL $0 \rightarrow w$	$R=\frac{1}{2}wL$	at $L/3$ from larger-intensity end
Trapezoid $w_1 \rightarrow w_2$	$R=(w_1+w_2)L/2$	locate by rectangle + triangle decomposition
Inclined point load	$F_x=F\cos\theta$; $F_y=F\sin\theta$	if θ measured from horizontal

Shear & Bending

Quantity / Case	Formula	Note / Unit
Shear force at a cut	$V = \text{algebraic } \Sigma \text{ transverse forces on one side}$	left or right side
Bending moment at a cut	$M = \text{algebraic } \Sigma \text{ moments of external forces on one side}$	about the cut
Reaction step	$\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$	before cutting beam

Ideal Pulley Principle

Same massless continuous rope + frictionless pulley $\Rightarrow$ same tension T throughout that rope.

A movable block supported by n rope segments has total upward support nT . Nested pulleys-এ level by level FBD করে tension relation বের করতে হবে.

Repeated Theory / QB

VERY HIGH

Define shear force and bending moment; find reactions, V and M at specified sections.

SF/BM definitions tagged: 2016,17,18,19,21,22

HIGH

Moment of force, couple and torque.

Set-note theory; recent paper includes torque/couple

HIGH

Pulley equilibrium - determine W / reaction.

Recurring: one supplied problem tagged 2017,18,20,22

HIGH

Convert UDL/UVL/trapezoid to equivalent point loads correctly.

Essential beam step

B2 SOLUTION ARCHITECTURE

SFBM & Pulley - Exam Workflow

Beam: 6-step method

1. Replace distributed loads by resultants for global reactions.
2. Draw full-beam FBD.
3. Find support reactions.
4. Cut at requested section.
5. Choose easier side and solve V.
6. Take moments about cut to solve M; cross-check other side if time.

Pulley: 5-step method

1. Mark each continuous rope separately.
2. Assign $T_1, T_2, \dots$ per rope.
3. FBD each movable pulley/block.
4. Use nT support relations from actual strands.
5. Finish with beam/lever moment equilibrium if pulley connects to beam.

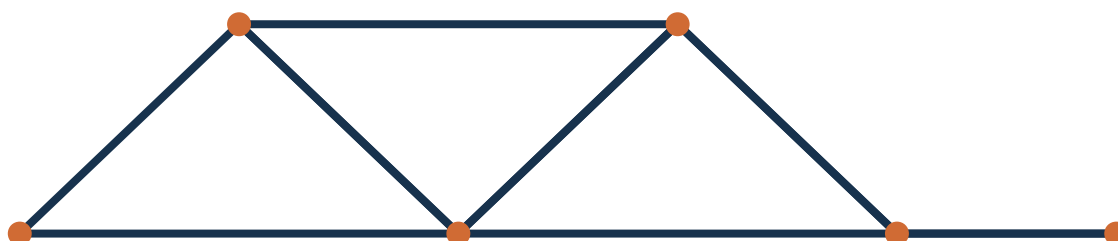
Pulley trap

"Each pulley halves the weight" মুখস্থ rule ব্যবহার করবেন না। It only happens in particular geometries. Count supporting rope segments from the actual figure.

B3

Truss · Plane / Rectilinear Motion

Two-force members + equilibrium at joints/sections



Pin-jointed triangular framework

Truss essentials

Item	Rule
Ideal truss	Pin-jointed straight members forming stable triangular framework.
Member action	Axial only: tension or compression under ideal assumptions.
Method of joints	At each joint: $\Sigma F_x=0$, $\Sigma F_y=0$.
Method of sections	Cut through desired members, then use $\Sigma F_x=0$, $\Sigma F_y=0$, $\Sigma M=0$ for one side.
Sign interpretation	Assume tension outward; negative result means compression.

Motion formulas

Quantity / Case	Formula	Note / Unit
Plane motion	all particle paths remain in one plane	definition
Rectilinear translation	straight parallel paths	prepared note framing
Projectile trajectory	$y=x \tan\theta - gx^2/(2u^2\cos^2\theta)$	particle model

QB targets

VERY HIGH

Compute external reactions and member forces in a truss; state T/C.

Repeated regular-paper archetype

HIGH

Define truss and state fundamental assumptions.

Prepared truss note: 2022 tag

HIGH

Draw different roof/bridge trusses (Pratt/Howe/Fink etc.).

Prepared note

HIGH

Method of joints and method of sections.

Prepared note explicitly uses both

MEDIUM

Define plane motion / rectilinear motion; projectile motorcycle problem.

Seen in regular papers and plane-motion note

B3 SHORTCUT

Truss - Fast but Safe Technique

Method of Joints

Unknown ≤ 2 এমন joint দিয়ে শুরু করুন. Member force initially **tension** ধরে joint থেকে outward arrow দিন. Result negative হলে compression.

Method of Sections

Specific 2-3 member চাইলে এমন section কাটুন যাতে ideally ≤ 3 unknown থাকে. তারপর এমন point-এ moment নিন যেখানে দুই unknown-এর lines intersect করে - এক equation-এ তৃতীয় member পাওয়া যায়.

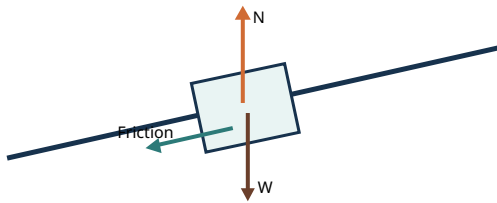
Source boundary

Zero-force member was explicitly asked in the 2016 QB and appears again in recent theory context, but the supplied set-wise truss note does not give a clean standalone rule sheet for it. তাই এই manual-এ unsupported memorized rule invent না করে only QB target হিসেবে retain করা হয়েছে.

B4

Friction · Belt · Wedge · Flexible Cord

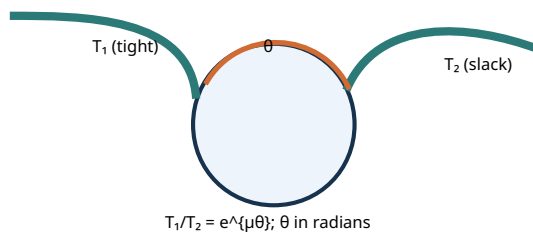
Impending motion + cable geometry



Quantity / Case	Formula	Note / Unit
Static friction	$0 \leq F_s \leq \mu_s N$	self-adjusting before limiting
Limiting friction	$F_{lim} = \mu_s N$	impending motion
Kinetic friction	$F_k = \mu_k N$	sliding
Angle of friction	$\tan \phi = \mu$	resultant reaction angle
Angle of repose	$\tan \alpha = \mu$	at impending sliding on incline

Sliding vs Tipping

Compute force required for **impending sliding** and force required for **impending tipping** separately. যে condition কম applied force-এ ঘটে, body প্রথমে সেটিই করবে.



Belt friction

$$T_1/T_2 = e^{\mu\theta}$$

T_1 = tight-side tension, T_2 = slack-side tension, θ = angle of contact in radians.

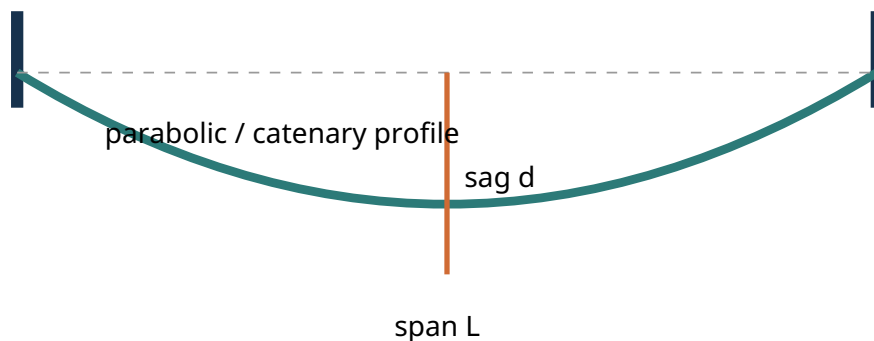
Total tangential frictional effect on the belt contact is related to $T_1 - T_2$.

Wedge

A wedge is a tapered solid driven/pushed between surfaces. At impending motion, replace each rough contact by a resultant reaction inclined by friction angle $\phi = \tan^{-1} \mu$ from the normal, in the direction opposing impending relative motion.

B4 FLEXIBLE CORD

Parabolic Cable vs Catenary



Parabolic chord / cable

Source condition: load uniformly distributed in the **horizontal direction**. A cable carrying mainly uniform horizontal load is modeled parabolically; the note also uses a parabola approximation for self-weight when sag is small (about <10% of span).

$$y = wx^2/(2H)$$

$$H = wL^2/(8d)$$

$$T = \sqrt{(H^2 + (wx)^2)}$$

For equal-level supports: $T_{\max} = \sqrt{H^2 + (wL/2)^2}$.

Catenary

When the cable's own weight is uniformly distributed per unit **actual cable length** and gravity is the loading, its exact curve is a catenary.

$$y = K \cosh(x/K)$$

$$s = K \sinh(x/K)$$

$$T = \sqrt{(H^2 + (ws)^2)}$$

Memory trick

Parabola = load per horizontal span. Catenary = self-weight per actual cable length. এই distinction theory question-এ আগে লিখলে answer পরিস্কার হয়.

QB targets

VERY HIGH

Distinguish angle of friction and angle of repose; state friction laws / slide-or-tip.

Repeated regular-paper pattern

VERY HIGH

For belt friction prove $T_1 = T_2 e^{\mu\theta}$.

Explicit recurring proof

VERY HIGH

Parabolic chord: derive $y = wx^2/(2H)$ and $H = wL^2/(8d)$.

2023/2025 style questions; supplied note derivation

HIGH

Conditions for parabola vs catenary; derive $y = K \cosh(x/K)$.

2015 QB explicitly asks catenary proof

HIGH

Wedge impending-motion problem.

Recent 2023-2025 pattern

Most Important Theory Questions - Priority Order

- 1 **FBD, transmissibility, two-force member, equilibrium** - definitions + correct sketch.
- 2 **Shear force and bending moment** - definitions, moment/couple/torque.
- 3 **Centroid, axis of symmetry, CG/CM relationship.**
- 4 **Rectangular MOI, polar MOI, radius of gyration** + applications.
- 5 **Angle of friction vs angle of repose; laws of friction.**
- 6 **Truss definition, assumptions, T/C concept, analysis methods.**
- 7 **Pappus-Guldinus two theorems** and meaning of generating curve/area.
- 8 **Product of inertia** - definition, sign, symmetry, application.
- 9 **Work, kinetic energy, power** and method selection.
- 10 **Linear impulse and momentum;** angular impulse-momentum.
- 11 **Parabolic chord vs catenary** - loading conditions.
- 12 **Support systems** - roller, pin, fixed reactions.
- 13 **Plane and rectilinear motion.**
- 14 **Wedge** and impending motion.

Study order

প্রথমে theory definitions 2-3 line করে নিখুঁত করুন; এরপর attached derivation; সবশেষে numerical pattern.
Definition + figure + governing equations দিলে theory answer অনেক শক্ত হয়.

Derivation Priority Ladder

নিচের derivation-গুলো exam-ready sequence-এ লেখা। “Given → differential relation → integrate/sum → result → condition” pattern অনুসরণ করুন।

1. Perpendicular-axis theorem: $J = I_x + I_y$

VERY HIGH

1. For an element dA at coordinates (x,y) , its distance from the perpendicular z -axis is r , where $r^2 = x^2 + y^2$.
2. Polar moment: $J = \int r^2 dA = \int (x^2 + y^2) dA$.
3. Split the integral: $J = \int x^2 dA + \int y^2 dA$.
4. By definitions, $\int x^2 dA = I_y$ and $\int y^2 dA = I_x$.

Final result: $J = I_x + I_y$

Condition: x and y lie in the plane, z is perpendicular, and all three axes pass through the same point.

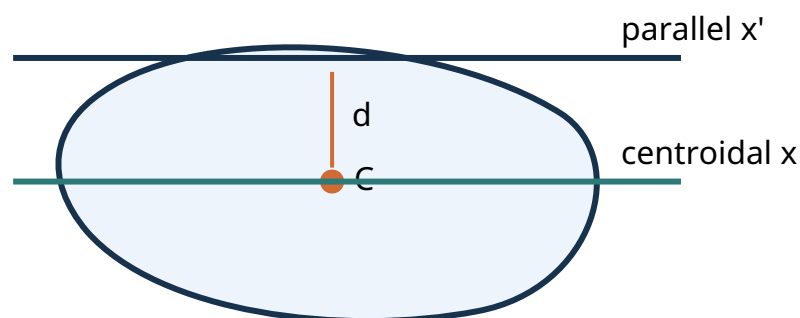
2. Parallel-axis theorem for area

VERY HIGH

1. Let x be a centroidal axis and x' a parallel axis at distance d .
2. For dA at coordinate y from centroidal axis, distance from x' is $(y+d)$ (sign choice does not affect final d^2 term).
3. $I_{x'} = \int (y+d)^2 dA = \int y^2 dA + 2d \int y dA + d^2 \int dA$.
4. Because x is centroidal, $\int y dA = 0$; and $\int dA = A$.
5. Therefore $I_{x'} = \bar{I}_x + Ad^2$.

Final result: $I = \bar{I} + Ad^2$

From $I = Ak^2$ and $\bar{I} = A \bar{k}^2$: $k^2 = \bar{k}^2 + d^2$.



Centroid & Standard Area MOI

3. Centroid of a triangle

HIGH

1. Consider a triangle of base b and height h . Take a horizontal strip at height y .
2. By similar triangles, strip width = $b(h-y)/h$, so $dA = [b(h-y)/h]dy$.
3. First moment about base: $\int y dA = (b/h) \int_0^h y(h-y)dy = bh^2/6$.
4. Total area $A = bh/2$.
5. $\bar{y} = (\int y dA)/A = (bh^2/6)/(bh/2) = h/3$.

Final result: $\bar{y} = h/3$ from the base

For the right-triangle orientation used in the supplied note, the corresponding horizontal coordinate is $\bar{x} = b/3$ from the adjacent vertical side.

4. Rectangle MOI about centroidal x-axis

HIGH

1. For rectangle width b , height h , choose horizontal strip $dA = b dy$.
2. $I_x = \int y^2 dA = b \int_{-h/2}^{h/2} y^2 dy$.
3. Symmetry: $I_x = 2b \int_0^{h/2} y^2 dy = 2b[y^3/3]_0^{h/2}$.
4. Simplify.

Final result: $I_x = bh^3/12$

Similarly $I_y = hb^3/12$.

5. Triangle MOI about base and centroidal parallel axis

HIGH

1. At height y , $dA = [b(h-y)/h]dy$.
2. $I_{\text{base}} = \int_0^h y^2 dA = (b/h) \int_0^h y^2(h-y)dy$.
3. Integrating gives $I_{\text{base}} = bh^3/12$.
4. Centroidal axis is $d = h/3$ from base. Use $I_{\text{base}} = \bar{I} + A d^2$.
5. $\bar{I} = bh^3/12 - (bh/2)(h/3)^2$.

Final result: $I_{\text{base}} = bh^3/12$; $\bar{I} = bh^3/36$

Product of Inertia & Pappus

6. Product-of-inertia transfer formula

HIGH

1. Let centroidal axes be $\bar{x}, \bar{y}$ and parallel reference axes be x, y , shifted by d_x, d_y .
2. For each dA : $x = \bar{x} + d_x$ and $y = \bar{y} + d_y$.
3. $P_{xy} = \int xy \, dA = \int (\bar{x} + d_x)(\bar{y} + d_y) dA$.
4. Expand: $\int \bar{x}\bar{y} dA + d_x \int \bar{y} dA + d_y \int \bar{x} dA + d_x d_y \int dA$.
5. Centroidal first moments vanish: $\int \bar{x} dA = \int \bar{y} dA = 0$.

Final result: $P_{xy} = \bar{P}_{xy} + A d_x d_y$

d_x and d_y carry signs; therefore POI can be positive or negative.

7. Pappus-Guldinus first theorem (surface area)

HIGH

1. Take a small curve element ds at distance y from the axis of revolution.
2. When revolved once, it sweeps approximately $dS = (2\pi y)ds$.
3. Integrate: $S = 2\pi \int y \, ds$.
4. For a curve, centroid coordinate satisfies $\bar{y} = (\int y \, ds) / L$, so $\int y \, ds = \bar{y}L$.

Final result: $S = 2\pi \bar{y} L$

8. Pappus-Guldinus second theorem (volume)

HIGH

1. Take a small area element dA at distance y from the axis.
2. On revolution it sweeps $dV = (2\pi y)dA$.
3. Integrate: $V = 2\pi \int y \, dA$.
4. Area centroid gives $\int y \, dA = \bar{y}A$.

Final result: $V = 2\pi \bar{y} A$

Mass MOI - Exam Proof Patterns

Editorial expansion

The supplied set note gives the standard mass-MOI formulas and the recent QB asks cylinder/sphere derivations. The step-by-step calculus below is an answer-ready expansion of those supplied formulas, not a replacement of the course notation.

9. Solid cylinder about its geometric (symmetry) axis

HIGH

1. Take a thin cylindrical ring of radius ρ , thickness $d\rho$ and length L . With uniform density ρ_m , $dm = \rho_m(2\pi\rho L)d\rho$.
2. $dI = \rho^2 dm = 2\pi\rho_m L \rho^3 d\rho$.
3. $I = 2\pi\rho_m L \int_0^R \rho^3 d\rho = (1/2)\pi\rho_m L R^4$.
4. Total mass $m = \rho_m \pi R^2 L$. Substitute.

Final result: $I = \frac{1}{2}mR^2$

10. Solid sphere about a diameter

HIGH

1. Use thin disks perpendicular to the diameter x . At coordinate x , disk radius $a = \sqrt{(R^2 - x^2)}$.
2. $dm = \rho_m \pi a^2 dx$. For a thin disk about its central normal (x -axis), $dI = (1/2)a^2 dm$.
3. $I = (\rho_m \pi / 2) \int_{-R}^R (R^2 - x^2)^2 dx$.
4. Integrating gives $I = (8\pi\rho_m R^5)/15$.
5. For sphere $m = \rho_m (4\pi R^3/3)$. Substitute.

Final result: $I = \frac{2}{5}mR^2$

For a tangent axis parallel to the diameter, apply parallel axis: $I_t = I_G + mR^2 = \frac{7}{5}mR^2$.

Work-Energy & Impulse-Momentum

11. Work-Kinetic Energy Principle

VERY HIGH

1. Newton: resultant force $R=ma$.
2. Along the path, $a=v(dv/ds)$, so $R ds = m v dv$.
3. Integrate from state 1 to 2: $\int_1^2 R ds = \int_{v_1}^{v_2} m v dv$.
4. Right side = $\frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$.

Final result: $U_{\text{net}} = KE_2 - KE_1 = \Delta KE$

For a system, net work is the algebraic sum of work of individual external forces. Normal reaction does zero work when displacement is tangent to a fixed smooth surface.

12. Linear Impulse-Momentum Principle

VERY HIGH

1. Newton: $\Sigma F = m dv/dt$ for constant mass.
2. Multiply by dt : $\Sigma F dt = m dv$.
3. Integrate from t_1 to t_2 : $\int \Sigma F dt = m \int dv$.
4. Right side = $mv_2 - mv_1$.

Final result: $\int_{t_1}^{t_2} \Sigma F dt = mv_2 - mv_1$

For constant resultant force: $R\Delta t = \Delta(mv)$.

13. Angular Impulse-Momentum (fixed axis)

HIGH

1. Rotational equation about fixed axis O: $\Sigma M_O = I_O \alpha$.
2. $\alpha = d\omega/dt$, so $\Sigma M_O dt = I_O d\omega$.
3. Integrate over the time interval.

Final result: $\int \Sigma M_O dt = I_O(\omega_2 - \omega_1)$

Kinematics & Projectile

14. Constant-acceleration displacement equation

MEDIUM

1. $a = dv/dt \Rightarrow$ integrate: $v = u + at$.
2. Also $v = ds/dt$. Substitute $v = u + at$.
3. Integrate $ds = (u + at)dt$ from 0 to t .

Final result: $s = ut + \frac{1}{2}at^2$

15. Projectile trajectory equation

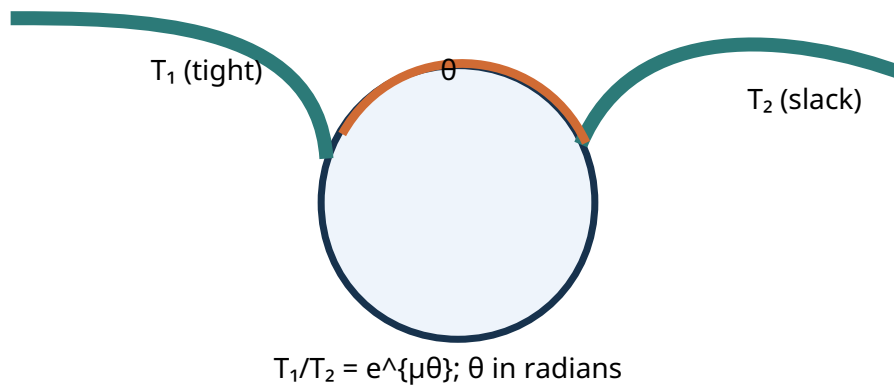
HIGH

1. Horizontal motion: $x = u \cos\theta \cdot t \Rightarrow t = x/(u \cos\theta)$.
2. Vertical motion: $y = u \sin\theta \cdot t - \frac{1}{2}gt^2$.
3. Substitute $t = x/(u \cos\theta)$.
4. Simplify the first term $u \sin\theta \cdot x/(u \cos\theta) = x \tan\theta$.

Final result: $y = x \tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

This is the core equation used in the supplied motorcycle-ditch problem pattern.

Belt Friction & Parabolic Cable



16. Belt friction relation

VERY HIGH

1. Consider a small belt element subtending $d\theta$. Let tension increase from T to $T+dT$ across the element.
2. For small $d\theta$, radial equilibrium gives $dN \approx T d\theta$.
3. At impending slip, elemental friction $dF = \mu dN$.
4. Tangential equilibrium gives $dT = dF = \mu T d\theta$.
5. $dT/T = \mu d\theta$. Integrate from T_2 to T_1 and 0 to θ .
6. $\ln(T_1/T_2) = \mu\theta$.

Final result: $T_1 = T_2 e^{\mu\theta}$

θ must be in radians; T_1 is tight side and T_2 slack side.

17. Parabolic cable equation under horizontal UDL

VERY HIGH

1. Consider cable from its lowest point O to a point at horizontal distance x . Let H be the constant horizontal component of tension.
2. Uniform load per horizontal length = w , so vertical load on segment = $w x$.
3. At the end, $T \sin\theta = wx$ and $T \cos\theta = H$, hence $\tan\theta = wx/H$.
4. Cable slope $dy/dx = \tan\theta = wx/H$.
5. Integrate with $y=0$ at $x=0$.

Final result: $y = wx^2/(2H)$

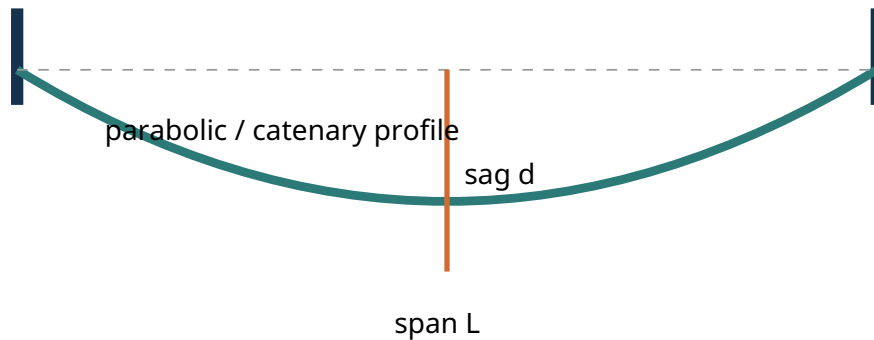
18. Horizontal tension from sag

VERY HIGH

1. For equal-level supports, lowest point is at midspan; at support $x=L/2$ and $y=d$.
2. Insert into $y=wx^2/(2H)$: $d=w(L/2)^2/(2H)$.
3. Solve for H .

Final result: $H = wL^2/(8d)$

Catenary - Source-Flow Derivation



19. Catenary equation $y = K \cosh(x/K)$

VERY HIGH

1. For a cable loaded only by its own weight per unit actual cable length w , consider a segment from the lowest point. Let s be arc length from the lowest point and H the constant horizontal component.
2. Vertical equilibrium gives $T \sin\theta = ws$; horizontal equilibrium gives $T \cos\theta = H$.
3. Therefore $\tan\theta = ws/H$. Let $K = H/w$, so $dy/dx = s/K$.
4. Arc-length relation: $ds^2 = dx^2 + dy^2 \Rightarrow ds/dx = \sqrt{1 + (dy/dx)^2} = \sqrt{1 + (s/K)^2}$.
5. Rearrange and integrate: $dx/K = ds/\sqrt{K^2 + s^2}$, giving $x/K = \sinh(s/K)$.
6. Hence $s = K \sinh(x/K)$.
7. Because $dy/dx = s/K = \sinh(x/K)$, integrate: $y = K \cosh(x/K) + C$.
8. If the origin is chosen at the lowest point with $y = K$ in the supplied source convention, $C = 0$. If $y = 0$ is preferred at the lowest point, use $y = K[\cosh(x/K) - 1]$.

Final result: $y = K \cosh(x/K)$ (source convention); $s = K \sinh(x/K)$

The coordinate origin convention must be stated. The supplied note presents $y = K \cosh(x/K)$ and later $y = K + d$ for support elevation.

One-Glance Proof / Derivation Checklist

Priority	Derivation	Anchor equation	Set
★★★★★	$J = I_x + I_y$	$r^2 = x^2 + y^2$	A2
★★★★★	Parallel-axis theorem	$I = \bar{I} + Ad^2$	A2
★★★★★	Work-energy	$\int R ds = \Delta(\frac{1}{2}mv^2)$	A4
★★★★★	Linear impulse-momentum	$\int F dt = \Delta(mv)$	A4
★★★★★	Belt friction	$dT/T = \mu d\theta$	B4
★★★★★	Parabolic cable	$dy/dx = wx/H$	B4
★★★★☆	Catenary	$dy/dx = s/K$	B4
★★★★☆	Triangle centroid	$\bar{y} = \int y dA/A$	A1
★★★★☆	Rectangle / triangle area MOI	$I = \int y^2 dA$	A2
★★★★☆	Product inertia transfer	$\int (\bar{x} + d_x)(\bar{y} + d_y) dA$	A3
★★★★☆	Pappus surface & volume	$dS = 2\pi y ds; dV = 2\pi y dA$	A3
★★★☆☆	Solid cylinder / sphere mass MOI	$I = \int r^2 dm$	A3
★★★☆☆	Projectile trajectory	eliminate t	A4/B3

Proof writing format

Exam-এ “Prove that” answer-এর শুরুতে figure/element + definition লিখুন, তারপর 4-6 clean equations, শেষে boxed result এবং condition. শুধু final formula লিখলে proof-এর marks পাওয়া কঠিন.

48-Hour Formula Sheet - Section A

Quantity / Case	Formula	Note / Unit
A1 Composite centroid	$\bar{x} = \Sigma Ax / \Sigma A; \bar{y} = \Sigma Ay / \Sigma A$	holes negative
A1 Triangle	$\bar{y} = h/3$	from base
A1 Sector	$\bar{r} = 2r \sin\alpha / (3\alpha)$	angle 2α
A1 Arc	$\bar{r} = r \sin\alpha / \alpha$	angle 2α
A2 Area MOI	$I_x = \int y^2 dA; I_y = \int x^2 dA$	L^4
A2 Polar	$J = I_x + I_y$	same point
A2 Transfer	$I = \bar{I} + Ad^2$	parallel axes
A2 Radius gyration	$k = \sqrt{I/A}$	length
A3 POI	$P_{xy} = \int xy \, dA$	signed
A3 POI transfer	$P = \bar{P} + Ad_x d_y$	signed shifts
A3 Principal	$I_{\max/\min} = (I_x + I_y)/2 \pm \sqrt{\{(I_x - I_y)/2\}^2 + P_{xy}^2}$	area
A3 Mass transfer	$I = \bar{I} + md^2$	mass
A3 Pappus	$S = 2\pi \bar{r} L; V = 2\pi \bar{r} A$	revolution
A4 Kinematics	$v = u + at; s = ut + \frac{1}{2}at^2; v^2 = u^2 + 2as$	const a
A4 Work-energy	$U_{\text{net}} = \Delta KE$	time-free
A4 Impulse	$\int F dt = \Delta(mv)$	short-time dynamics
A4 Projectile	$y = x \tan\theta - gx^2 / (2u^2 \cos^2\theta)$	no air

48-Hour Formula Sheet - Section B

Quantity / Case	Formula	Note / Unit
B1 Components	$F_x = F \cos \theta$; $F_y = F \sin \theta$	θ from x
B1 Resultant	$R = \sqrt{(\sum F_x)^2 + (\sum F_y)^2}$	direction by atan2
B1 Moment	$M = Fd_{\perp}$	N·m or lb·ft
B1 Equilibrium	$\sum F_x = 0$; $\sum F_y = 0$; $\sum M = 0$	2D rigid body
B2 UDL	$R = wL$ @ $L/2$	equivalent
B2 triangular UVL	$R = wL/2$ @ $L/3$ from larger end	equivalent
B2 Pulley	same rope $\Rightarrow$ same T; movable block support $= \sum T$	count strands
B3 Joint	$\sum F_x = 0$; $\sum F_y = 0$	truss
B3 Section	$\sum F_x = 0$; $\sum F_y = 0$; $\sum M = 0$	truss
B4 Limiting friction	$F = \mu N$	impending
B4 Friction angle	$\tan \phi = \mu$	also repose at limit
B4 Belt	$T_1/T_2 = e^{\mu \theta}$	θ rad
B4 Parabola	$y = wx^2/(2H)$; $H = wL^2/(8d)$	horizontal UDL
B4 Cable support T	$T_{\max} = \sqrt{[H^2 + (wL/2)^2]}$	equal-level supports
B4 Catenary	$y = K \cosh(x/K)$; $s = K \sinh(x/K)$	self-weight

Common Errors That Lose Marks

Geometry / MOI

- Sector এবং arc centroid formula mix করা.
- Triangle $h/3$ কোন side থেকে - reference না লেখা.
- Area MOI vs mass MOI unit mix.
- Parallel-axis-এ d centroid-to-axis distance না নেওয়া.
- POI dx,dy sign ignore করা.

Statics / Dynamics

- FBD-তে internal force accidentally show করা.
- Smooth contact reaction tangent এ আঁকা.
- UVL resultant location উল্টো নেওয়া.
- Pulley-তে tension ভাগ "guess" করা.
- Static friction always μN ধরা.
- Projectile reference elevation ভুল.

Minimum safe preparation if time is short

প্রতি section-এ ৩টি strong set choose করুন, কিন্তু ৪র্থ set-এর **definitions + formula + at least one proof** backup হিসেবে রাখুন। Question mix হলে এই backup marks বাঁচাবে.

Primary supplied-source map used

Section A: 1 - Centroid - by 2223007; Introduction/CG centroid slides; Centroids of Common Shapes; Moment of Inertia of Area slides; Product of Inertia by 2223007; Moment of Inertia of Masses by 2223007; Pappus-Guldinus slides; Dynamics by 2223007; Work, Power and Energy; Impulse Momentum.

Section B: 10-EQU~1 (FBD/equilibrium/supports); Shear Force and Bending Moment; Pulley by 2223007; Truss by 2223012; Plane Motion; Friction; Flexible Cord; Kabbo Sir moment/parallel coplanar forces.

Question-bank basis: Regular papers 2015-2021, 2023, 2024, 2025. Backlog is not used to inflate regular-exam repeat frequency.

Prepared as an exam-focused synthesis of the supplied course materials. Credit: Made by 2223033.